LaTeX – Typesetting Text

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UP FAMNIT

Line Breaking and Page Breaking

Justified Paragraphs

- LaTeX inserts the necessary line breaks and spaces between words by optimizing the contents of a whole paragraph.
- How the paragraphs are typeset depends on the document class.
- LaTeX always tries to produce the best line breaks possible.

Justified Paragraphs

In special cases it might be necessary to order LaTeX to break a line:

- To start a new line without starting a new paragraph use `\\` or `\newline`.
- To start a new page use `\newpage`.

Ready-Made Strings

Some very simple LaTeX commands for typesetting special text strings:

- For current date use

```
\today
```

- For your favorite typesetter use

```
\TeX{}
```

- For the name of the game use

```
\LaTeX{}
```

Special Characters and Symbols

Quotation Marks

- In LaTeX, use two ‘ (grave accent) for opening quotation marks and two ’ (vertical quote) for closing quotation marks.
- For single quotes you use just one of each.

```
``Please press the `x' key.''
```

“Please press the ‘x’ key.”

Slash (/) and Tilde (∼)

- To generate slash in LaTeX use `/` or `\slash`.
- To generate tilde in LaTeX use `\~{}` or `$\sim$`.

```
http://www.rich.edu/\~{}bush \\
http://www.clever.edu\slash$\sim$demo
```

http://www.rich.edu/~bush
http://www.clever.edu/∼demo

Ellipsis (...)

- Entering 'ellipsis' by just typing three dots would produce the wrong result.
- Instead, there is a special command for these dots: `\ldots`.

```
Not like this ... but like this:\\
New York, Tokyo, Budapest, \ldots
```

Not like this ... but like this:
New York, Tokyo, Budapest, ...

Accents and Special Characters

- LaTeX supports the use of accents and special characters from many languages.
- To place an accent on top of an i or a j, its dots have to be removed.
- This is accomplished by typing `\i` and `\j`.

```
H\^otel, na\"\i ve, sm\o rrebr\o d,
Sch\"onbrunner Schlo\ss{} Stra\ss e
```

Hôtel, naïve, smørrebrød, Schönbrunner Schloß Straße

Accents and Special Characters

- All sorts of accents being applied to the letter o.
- Naturally other letters work too.

ò	\‘o	ó	\’o	ô	\^o	õ	\~o
ō	\=o	ȯ	\.o	ö	\"o	ç	\c c
ŏ	\u o	ȏ	\v o	ő	\H o	ǫ	\c o
ọ	\d o	ọ̲	\b o	ōo	\t oo		
œ	\oe	Œ	\OE	æ	\ae	Æ	\AE
å	\aa	Å	\AA				
ø	\o	Ø	\O	ł	\l	Ł	\L
ı	\i	ȷ	\j	¡	!‘	¿	?‘

International Language Support

Babel Package

When you write documents in languages other than English, there are three areas where LaTeX has to be configured appropriately:

- All automatically generated text strings have to be adapted to the new language.
- LaTeX needs to know the hyphenation rules for the new language.
- Language specific typographic rules.

Activate the **babel** package by adding the command:

```
\usepackage[language]{babel}
```

Multiple Languages

- If you call **babel** with multiple languages

```
\usepackage[languageA,languageB]{babel}
```

then the last language in the option list will be active (i.e. languageB).

- Use the command

```
\selectlanguage{languageA}
```

to change the active language.

Titles, Chapters, and Sections

Chapters and Sections

- To help the reader find a way through your work, you should divide it into chapters, sections, and subsections.
- The following sectioning commands are available for the article class:

```
\section{...}
\subsection{...}
\subsubsection{...}
\paragraph{...}
\subparagraph{...}
```

Chapters and Sections

- When you work with the `report` or `book` class, an additional top-level sectioning command becomes available

```
\chapter{...}
```

- The `article` class does not know about chapters.
- The spacing between sections, the numbering and the font size of the titles will be set automatically by LaTeX.

Table of Contents

- LaTeX creates a table of contents by taking the section headings and page numbers from the last compile cycle of the document.
- The command

```
\tableofcontents
```

  expands to a table of contents at the place it is issued.
- A new document has to be compiled twice to get a correct table of contents.

Title

- The title of the whole document is generated by issuing a

```
\maketitle
```

  command.
- The contents of the title have to be defined by the commands

```
\title{...}, \author{...} and optionally
\date{...}
```

  before calling `\maketitle`.
- In the argument to `\author`, you can supply several names separated by `\and` commands.

Cross References

Cross References

- In books, reports and articles, there are often cross-references to figures, tables and special segments of text.
- LaTeX provides the following commands for cross referencing:

```
\label{marker}, \ref{marker} and
\pageref{marker}
```

Cross References

```
\label{marker}, \ref{marker} and
\pageref{marker}
```

- *Marker* is an identifier chosen by the user.
- LaTeX replaces `\ref` by the number of the section, subsection, figure, table, or theorem after which the corresponding `\label` command was issued.
- `\pageref` prints the page number of the page where the `\label` command occurred.

Cross References - Example

```
Cross References \label{sec:this}

A reference to this subsection looks like:
see section~\ref{sec:this} on
page~\pageref{sec:this}.
```

A reference to this subsection looks like: see section 5 on page 20.

Footnotes

Footnotes

- With the command

```
\footnote{footnote text}
```

  a footnote is printed at the foot of the current page.
- Footnotes should always be put after the word or sentence they refer to.
- Footnotes referring to a sentence or part of it should therefore be put after the comma or period.

Footnotes - Example

```
Footnotes\footnote{This is a footnote.}
are often used by people using \LaTeX.
```

Footnotes[a] are often used by people using LaTeX.

[a] This is a footnote.

Emphasized Words

Emphasized Words

- We can underline important words with the command

```
\underline{text}
```

- In printed books important words are emphasized by typesetting them in an *italic* font.

Emphasized Words

Use the command

```
\emph{text}
```

to emphasize text.

```
\emph{This text is emphasized}
and this is not.
```

This text is emphasized and this is not.

Environments

Environments

To call an environment in LaTeXuse

```
\begin{environment}
some text
\end{environment}
```

where `environment` is the name of the environment.

Environments

Environments can be nested within each other as long as the correct nesting order is maintained.

```
\begin{aaa}...
\begin{bbb}...
\end{bbb}...
\end{aaa}
```

Itemize, Enumerate, and Description

```
\begin{enumerate}
\item You can nest the list environments to your
taste:
\begin{itemize}
\item But it might start to look silly.
\item[-] With a dash.
\end{itemize}
\item Therefore remember:
\begin{description}
\item[Stupid] things will not become smart because
they are in a list.
\item[Smart] things, though, can be presented
beautifully in a list.
\end{description}
\end{enumerate}
```

Itemize, Enumerate, and Description

1. You can nest the list environments to your taste:
   - But it might start to look silly.
   - With a dash.
2. Therefore remember:

   Stupid things will not become smart because they are in a list.

   Smart things, though, can be presented beautifully in a list.

Itemize, Enumerate, and Description

- The **itemize** environment is suitable for simple lists.
- The **enumerate** environment is suitable for enumerated lists.
- The **description** environment is suitable for descriptions.

Flushleft, Flushright, and Center

- The environments **flushleft** and **flushright** generate paragraphs that are either left- or right-aligned.
- The **center** environment generates centred text.
- If you do not issue \\ to specify line breaks, LaTeX will automatically determine line breaks.

Flushleft - Example

```
\begin{flushleft}
This text is\\ left-aligned. \LaTeX{} is
not trying to make each line the same
length \ldots
\end{flushleft}
```

This text is
left-aligned. LaTeX is not trying to make each line
the same length ...

Flushright - Example

```
\begin{flushright}
This text is right-\\aligned. Again
\LaTeX{} is not trying to make each
line the same length \ldots
\end{flushright}
```

This text is right-
aligned. Again LaTeX is not trying to make each line
the same length ...

Center - Example

```
\begin{center}
At the centre\\of the earth.
\end{center}
```

At the centre
of the earth.

Abstract

- In scientific publications it is customary to start with an abstract which gives the reader a quick overview of what to expect.
- LaTeX provides the **abstract** environment for this purpose.
- Normally abstract is used in documents typeset with the article document class.

Abstract - Example

```
\begin{abstract}
A quick overview of what to expect.
\end{abstract}
```

Abstract

A quick overview of what to expect.

Printing Verbatim

Text that is enclosed between

```
\begin{verbatim}
```

and

```
\end{verbatim}
```

will be directly printed with all line breaks and spaces, without any LaTeX command being executed.

Printing Verbatim

- Within a paragraph, similar behavior can be accessed with

  `\verb+text+`

- The + is just an example of a delimiter character.
- Use any character except letters, * or space.
- The **verbatim** environment and the verb command may not be used within parameters of other commands.

Printing Verbatim - Example

```
The \verb+\ldots+ command \ldots

\begin{verbatim}
10 PRINT "   HELLO WORLD   ";
20     GOTO     10
end{verbatim}
```

The \ldots command ...

```
10 PRINT "   HELLO WORLD   ";
20     GOTO     10
```

Tabular

- The **tabular** environment is used for tables with optional horizontal and vertical lines.
- LaTeX determines the width of the columns automatically.

Tabular

```
\begin{tabular}[pos]{table spec} ... \end{tabular}
```

The table spec argument of the command defines the format of the table:

- l for a column of left-aligned text,
- r for right-aligned text,
- c for centred text,
- | for a vertical line.

Tabular

```
\begin{tabular}[pos]{table spec} ... \end{tabular}
```

- The pos argument specifies the vertical position of the table relative to the baseline of the surrounding text.
- Use one of the letters t, b and c to specify table alignment at the top, bottom or centre.

Tabular

- Within a **tabular** environment:
  `&` jumps to the next column,
  `\\` starts a new line and
  `\hline` inserts a horizontal line.
- Add partial lines by using `\cline{i-j}`, where `i` and `j` are the column numbers the line should extend over.
- Join columns by using the `\multicolumn` command.

Tabular - Example

```
\begin{tabular}{|r|l|}
\hline
7C0 & hexadecimal \\
3700 & octal \\ \cline{2-2}
11111000000 & binary \\
\hline \hline
1984 & decimal \\
\hline
\end{tabular}
```

7C0	hexadecimal
3700	octal
11111000000	binary
1984	decimal

Tabular - Example

```
\begin{tabular}{|c|c|}
\hline
\multicolumn{2}{|c|}{Ene} \\
\hline
Mene & Muh! \\
\hline
\end{tabular}
```

Ene	
Mene	Muh!

Floating Bodies

Floating Bodies

- LaTeX offers two environments for floating bodies; one for figures and one for tables.
- Any material enclosed in a **figure** or **table** environment will be treated as floating matter.

```
\begin{figure}[placement specifier]...\end{figure}
```

```
\begin{table}[placement specifier]...\end{table}
```

- A placement specifier is used to tell LaTeX about the locations to which the float is allowed to be moved.

Floating Bodies

A placement specifier is constructed by building a string of float-placing permissions:

Spec	Permission to place the float ...
`h`	at the very place in the text where it occurred (here). Useful for small floats.
`t`	at the top of a page
`b`	at the bottom of a page
`p`	on a special page containing only floats.
`!`	without considering most of the internal parameters, which could otherwise stop this float from being placed.

Floating Bodies

- A table could be started with the following line:

```
\begin{table}[!hbp]
```

- The placement specifier `[!hbp]` allows LaTeX to place the table right here (`h`) or at the bottom (`b`) of some page or on a special floats page (`p`), and all this even if it does not look that good (`!`).
- If no placement specifier is given, the standard classes assume `[tbp]`.

Floating Bodies

- There are some more things to mention about the **table** and **figure** environments.
- Use the

```
\caption{caption text}
```

  command to define a caption for the float.
- A running number and the string "Figure" or "Table" will be added by LaTeX.

Floating Bodies

- The two commands `\listoffigures` and `\listoftables` print a list of figures or list of tables, respectively (like `\tableofcontents`).
- Use `\label` and `\ref` to create a reference to a float within your text.
- Note that the `\label` command must come after the `\caption` command since you want it to reference the number of the caption.

Floating Bodies - Example

```
\begin{figure}[!hbtp]
\makebox[\textwidth]{\framebox[5cm]
{\rule{0pt}{5cm}}}
\caption{Five by Five in Centimetres.
\label{white}}
\end{figure}
Figure~\ref{white} is an example of
Pop-Art.
```

Floating Bodies - Example

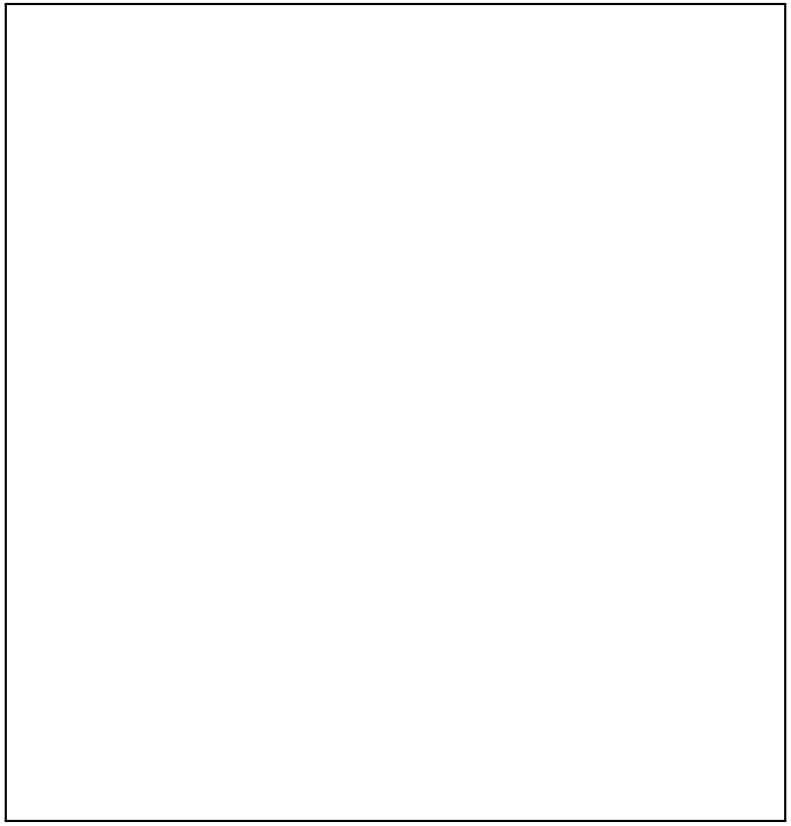

Figure : Five by Five in Centimetres.

Figure 1 is an example of Pop-Art.

Floating Bodies

- Command `\clearpage` orders LaTeX to immediately place all floats remaining in the queues and then start a new page.
- Command `\cleardoublepage` even goes to a new right-hand page.

Bibliography

Bibliography

- Produce a bibliography with the **thebibliography** environment.
- Each entry starts with

```
\bibitem{marker}
```

- The marker is then used to cite the book, article or paper within the document:

```
\cite{marker}
```

Bibliography

- The parameter after the `\begin{thebibliography}` command defines how much space to reserve for the number of entries.
- In next example, `{99}` tells LaTeX to expect that none of the bibliography item numbers will be wider than the number 99.

Bibliography - Example

```
Partl~\cite{pa} has
proposed that \ldots
\begin{thebibliography}{99}
\bibitem{pa} H.~Partl:
\emph{German \TeX},
TUGboat Volume~9, Issue~1 (1988)
\end{thebibliography}
```

Partl [1] has proposed that ...

H. Partl: *German TEX*, TUGboat Volume 9, Issue 1 (1988)

Indexing

Indexing

- With LaTeX and the support program **makeindex** an index can be generated.
- To enable their indexing feature of LaTeX, the **makeidx** package must be loaded in the preamble and the special indexing commands must be enabled by putting the `\makeindex` command in the preamble.

Indexing

- The content of the index is specified with

```
\index{key@formatted_entry}
```

  commands, where formatted_entry will appear in the index and key will be used for sorting.
- You enter the index commands at the points in the text that you want the final index entries to point to.

Indexing

Table explains the syntax with several examples.

Example	Index Entry	Comment
`\index{hello}`	hello, 1	Plain entry
`\index{hello!Peter}`	Peter, 3	Subentry under 'hello'
`\index{Sam@\textsl{Sam}}`	*Sam*, 2	Formatted entry
`\index{Lin@\textbf{Lin}}`	**Lin**, 7	Formatted entry
`\index{Jenny|textbf}`	Jenny, **3**	Formatted page number
`\index{Joe|textit}`	Joe, *5*	Formatted page number

Indexing

- When the input file is processed with LaTeX, each `\index` command writes an appropriate index entry, together with the current page number, to a special file.
- The file has the same name as the LaTeX input file, but a different extension (`.idx`).
- This .idx file can then be processed with the **makeindex** program.

Indexing

- The **makeindex** program generates a sorted index with the same base file name, but this time with the extension .ind.
- If now the LaTeX input file is processed again, this sorted index gets included into the document at the point where LaTeX finds `\printindex`.
- The **showidx** package prints out all index entries in the left margin of the text.